

A Novel Modified Triangular Shaped Microstrip Patch Antenna for C-band Applications in Wireless and Satellite Communication

Sayed Shifat Ahmed
Department of Electrical and
Electronic Engineering
RTM Al-Kabir Technical University
Sylhet-3100, Bangladesh
shikor99@gmail.com

Shah Md Toufik Ahmed
Department of Electrical and Electronic
Engineering
Rajshahi Science & Technology University
Natore-6260, Bangladesh
toufik14.eece@gmail.com

Md Arafath Hossain Rafi
Department of Electrical and
Electronic Engineering
East West University
Dhaka-1212, Bangladesh
2017-2-80-032@std.ewubd.edu

Md. Mehedi Hassan
Department of Electrical and
Electronic Engineering
East West University
Dhaka-1212, Bangladesh
hassan.eee18@gmail.com

Md. Saffat Hossain
Department of Electrical and
Electronic Engineering
East West University
Dhaka-1212, Bangladesh
saffat627@gmail.com

Sadia Tabassum
Department of Electrical and
Electronic Engineering
East West University
Dhaka-1212, Bangladesh
2017-3-50-012@std.ewubd.edu

Abstract—This paper introduces a novel triangular-shaped microstrip patch antenna designed for high-performance communication systems, specifically in the C-band (5.58 GHz to 8.72 GHz). The antenna, performed on Rogers RT5880 substrate, addresses significant challenges in traditional designs, including limited bandwidth and low efficiency, by optimizing the patch geometry and feed configuration. Using CST Microwave Studio for simulation and analysis, the antenna achieves a 3140 MHz bandwidth, 95.52% efficiency, and superior impedance matching and is well-suited for satellite and 5G communication systems that require high data rates and low latency. The surface current distribution highlights efficient radiation, particularly along the patch edges, ensuring reliable signal transmission. The results confirm that this antenna provides a robust solution to meet the growing demands of next-generation communication networks, offering enhanced performance compared to conventional designs.

Keywords—Triangular patch antenna, Rogers RT5880, C-band applications, satellite communication.

I. INTRODUCTION

The rapid growth of wireless communication and satellite systems has intensified the demand for high-performance antennas supporting high data rates and low-latency links [1]. Microstrip patch antennas (MPAs) are widely used due to their low profile, cost-effectiveness, and ease of integration, particularly in the C-band (4–8 GHz) [2]. However, conventional MPAs face challenges including narrow bandwidth, low efficiency, and poor impedance matching, limiting their effectiveness for high-speed, data-intensive applications [3]. High-gain, frequency-agile antennas are critical for satellite payload interfaces and LEO/HTS architectures [4]. Despite advancements, most MPAs fail to meet the >90% efficiency required for high-power satellite communication [5], necessitating novel designs that enhance bandwidth, efficiency, and impedance performance for modern wireless and satellite systems.

Recent research emphasizes compact, high-gain antennas for C- and X-band satellite, radar, and high-frequency wireless applications. Performance enhancement techniques include

notch loading [6], parasitic elements [7], and novel patch geometries [8]. Abdullah-Al-Mamun et al. [9] employed PBG substrates and DGS, achieving -38.7 dB return loss and 9.08 dBi gain. Dual-band designs using flipped Z-shaped slots on Rogers RO5880 substrates demonstrated return losses of $-68.3/-33.2$ dB and gains of 7.01/5.31 dBi [10], while other approaches achieved similar high-efficiency, dual-band performance [11]. These strategies, often optimized via HFSS, advance the development of compact, broadband, and high-gain antennas for modern communication systems.

The current limitations of traditional microstrip antennas, particularly in terms of bandwidth and efficiency, have been a major hurdle for the effective deployment of satellite and wireless communications. This design makes a significant contribution to the field by proposing a modified triangular microstrip patch antenna with the following key improvements: a bandwidth of 3.14 GHz, a resonance frequency of 8.26 GHz, 95.52% efficiency, and VSWR of 1.01 to provide a broadband solution in C-band spectrum. The compact design ensures effective impedance matching, broadband performance, and high efficiency, making it suitable for satellite communication, 5G, and other high-speed, low-latency wireless applications.

II. DESIGN ANALYSIS

A compact triangular microstrip patch antenna for C-band (5.58–8.72 GHz) satellite communication is presented, implemented on a Rogers RT5880 substrate ($\epsilon_r = 2.2$, 0.79 mm thickness, 25×25 mm²) with copper cladding on both faces. Substrate thickness directly affects resonant frequency and bandwidth and the triangular patch shape with partial ground plane (5.7 mm) and geometrical modifications, including an inner triangular arm and diagonal trimming, is optimized to resonate at 8.26 GHz. Excited via a 50 Ω microstrip feed, the antenna achieves a 3.14 GHz bandwidth, high radiation efficiency, and low return loss. CST simulations confirm stable reflection, efficient power transfer, and wideband operation, demonstrating compact, high-performance design suitable for high-speed, data-intensive satellite communication systems.

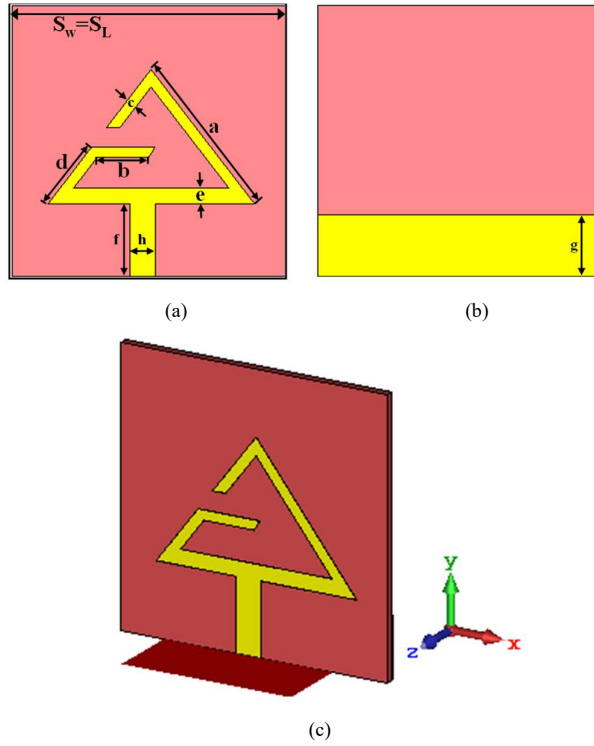


Fig. 1. Proposed MPA (a) front view, (b) back view and (c) 3D view.

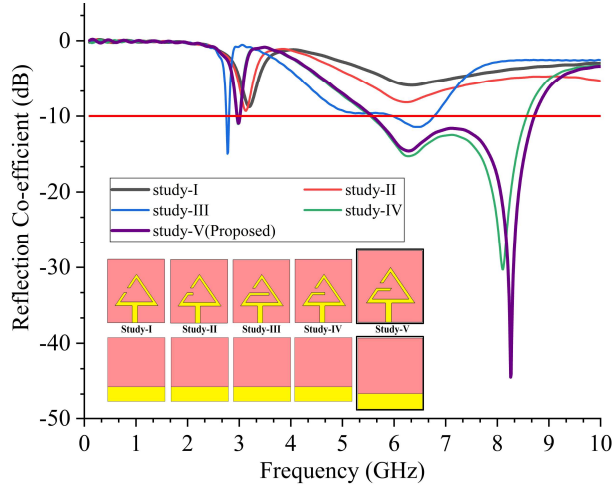


Fig. 2. Case study of varying arm length.

TABLE I: PHYSICAL DIMENSIONS OF PROPOSED ANTENNA

Parameter	Dimension (mm)	Parameter	Dimension (mm)
Substrate width, length ($S_w=S_L$)	25	Triangular lower left arm (d)	6.53
Triangular arm length (a)	15.59	Feeder width (h)	2.30
Triangular inner arm (b)	5.80	Feeder height (f)	6.70
Triangular upper arms width (c)	0.97	Ground height (g)	5.70
Triangular base arm width (e)	1.40	Inner arm angle (α)	127.10°

III. ANTENNA PERFORMANCE EVALUATION

A. Reflection Co-efficient

The return loss (S_{11}) is a key parameter for evaluating impedance matching in antennas. For the proposed triangular microstrip patch antenna, S_{11} was analyzed over 5.58–8.72 GHz, with resonance at 8.26 GHz. The measured return loss of -44.44 dB demonstrates excellent matching, minimal reflection, and efficient power transfer. The corresponding reflection coefficient confirms effective radiation within the C-band. These results validate the antenna's capability to support high-performance satellite communication and high-speed wireless networks with stable transmission and reception characteristics.

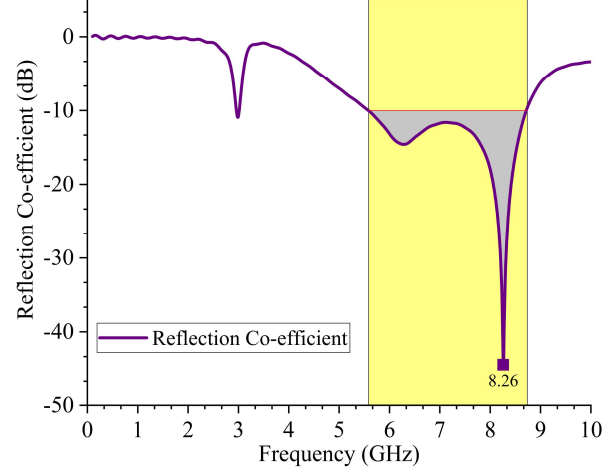


Fig. 3. Reflection Co-efficient curve for proposed MPA.

B. Impedance Matching

The proposed antenna exhibits excellent impedance matching, achieving a VSWR of 1.01, significantly below the 2:1 threshold for efficient operation. This value confirms minimal reflection and near-ideal power transfer with the 50Ω feed line, ensuring stable communication performance across the operating band (Fig. 4). The calculated input impedance of $50 - j0.57 \Omega$ (Fig. 5) indicates negligible reactance, validating effective matching to the reference impedance. Such performance highlights the antenna's suitability for modern high-frequency satellite communication applications.

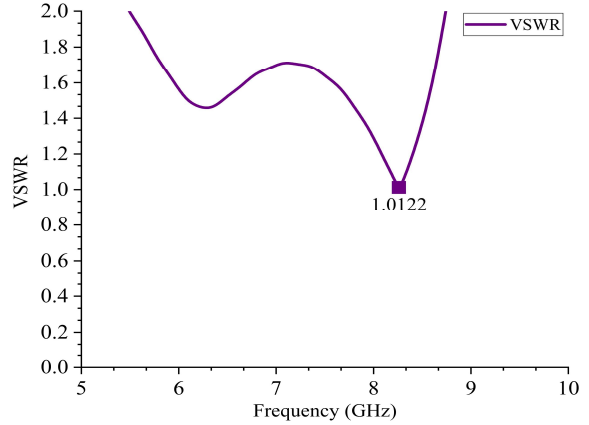


Fig. 4. VSWR curve for proposed MPA.

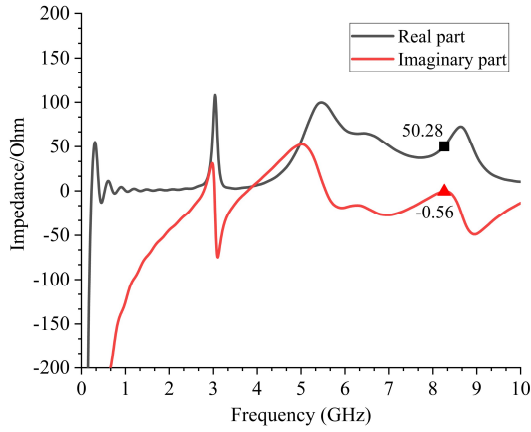


Fig. 5. Z-parameter of the proposed MPA.

C. Radiation Efficiency

The radiation efficiency of an antenna is a critical parameter that defines the proportion of power effectively radiated into free space as compared to the total power fed into the antenna. As shown in Fig. 6, the proposed antenna demonstrates a radiation efficiency of 95.52%, which is quite high for microstrip antennas. This result indicates that only a small fraction of power is lost due to factors such as material losses, surface resistance, and other inefficiencies typically associated with microstrip designs. A high radiation efficiency is crucial for satellite communication systems, where antenna efficiency directly impacts the system's overall performance, especially for high-power communication links in space. The design optimization, which includes the selection of Rogers RT5880 substrate with low loss properties and the precise dimensioning of the patch, has contributed to achieving this high efficiency. The simulation results suggest that the antenna exhibits minimal conductive and dielectric losses, leading to effective power radiated in the desired frequency range.

D. Directivity and Gain

The antenna exhibits a measured gain of 4.69 dB and calculated directivity of 4.965 dBi at the 8.26 GHz resonance (Figs. 7–8). The gain indicates efficient radiation along the direction of maximum power, while the directivity confirms effective energy concentration in desired directions. The 3D radiation pattern demonstrates focused C-band transmission, making the design suitable for satellite communication and point-to-point wireless links, ensuring high signal strength and directional efficiency.

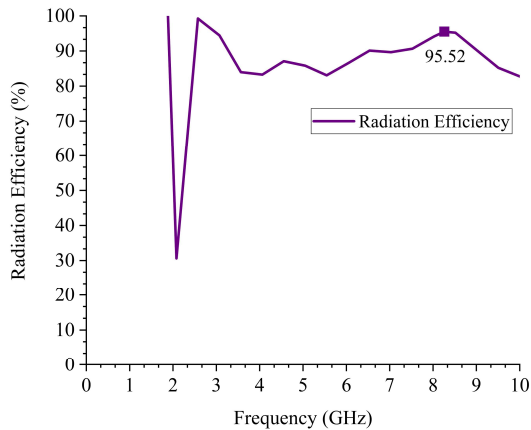


Fig. 6. Radiation efficiency of the proposed MPA.

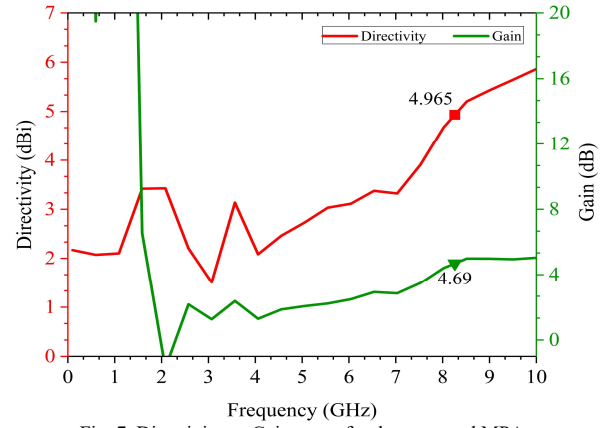


Fig. 7. Directivity vs Gain curve for the proposed MPA.

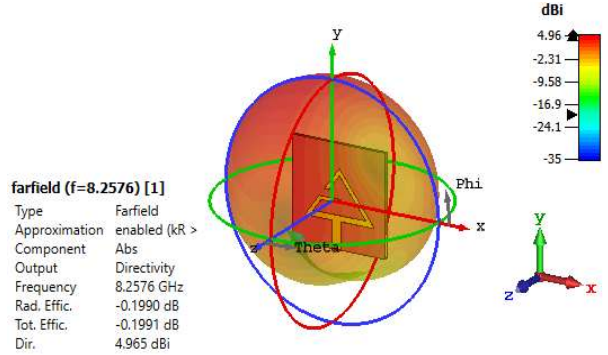


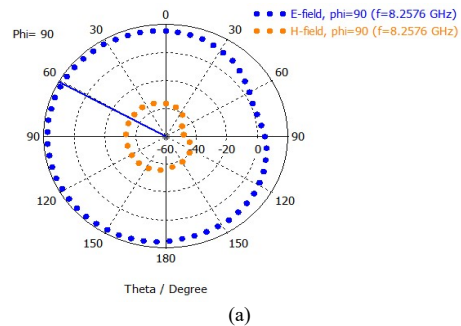
Fig. 8. Radiation pattern in 3D view for the proposed MPA at 8.2576 GHz frequency.

E. Radiation Characteristics

The E-field and H-field distributions of the proposed triangular microstrip patch antenna at 8.2576 GHz reveal its radiation characteristics. At $\phi = 90^\circ$, the E-field peaks at 17.6 dB(V/m) with a -3 dB angular width of 188.1° , while the H-field reaches -33.9 dB(A/m). At $\phi = 0^\circ$, the E-field and H-field peak at 15.3 dB(V/m) and -36.2 dB(A/m), respectively, with a 99.4° beamwidth. The aligned E- and H-field patterns confirm linearly polarized, directional radiation with wide angular coverage, ensuring efficient energy delivery for high-performance satellite and wireless communication systems.

F. Surface Current Distribution Analysis

The surface current distribution of the proposed triangular microstrip patch antenna at resonance shows maximum intensity (146 A/m) near the feed point, concentrated along the inner diagonal arm and patch edges. The current gradually decreases toward the triangle tip, illustrating typical microstrip patch behavior and confirming effective excitation and radiation across the structure.



(a)

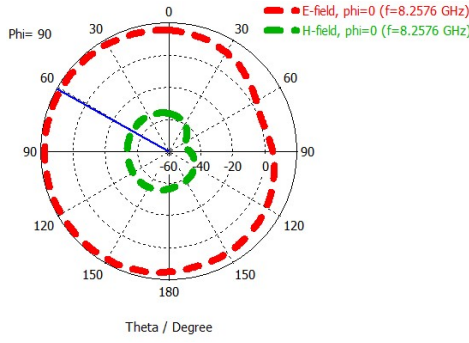


Fig. 9. E, H-field analysis at (a) $\phi=90$ and (b) $\phi=0$ for the proposed MPA.

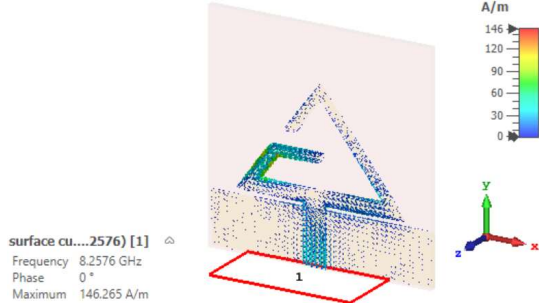


Fig. 10. Surface current distribution of the proposed MPA.

TABLE II: PERFORMANCE METRICS OF THE PROPOSED ANTENNA

Parameter Name	Value
Frequency Range (GHz)	5.58–8.72
Bandwidth (MHz)	3140
Centre Frequency (GHz)	8.26
Return loss (dB)	−44.444
Directivity (dBi)	4.965
Gain (dB)	4.69
Maximum Radiation Efficiency (%)	95.52
VSWR	1.01
Impedance (Ω)	50−j0.57

IV. COMPARATIVE ANALYSIS

As shown in Table III, the proposed triangular-shaped microstrip patch antenna outperforms traditional designs in terms of bandwidth, efficiency, and impedance matching, addressing key limitations of conventional antennas and making it ideal for next-generation communication systems. Rogers RT5880 was selected for its low dielectric loss and stability ($\epsilon_r \approx 2.2$), which aided achieving the reported high radiation efficiency (95.52%). Lower-cost alternatives such as standard FR-4 (higher loss) or mid-range low-loss laminates typically degrade radiation efficiency.

TABLE III: COMPARISON OF THE PROPOSED ANTENNA PERFORMANCE WITH EXISTING DESIGNS

Ref.	[6]	[7]	[8]	[9]	[10]	[11]	<i>Proposed Work</i>
Size (L×W×H) mm ³	35×25×0.79	32×32×3.07	25×25×1.6	44×41×1.6	41×34×1.6	34×40×1.5	25×25×0.79
Substrate Material	Polyamide	Taconic RF-35	FR4 Epoxy	Rogers RT5880	Rogers RO5880	FR4	Rogers RT5880
Operating Freq. Range (GHz)	6.58 – 8.54	5.15 – 6.30	5.0 – 7.0	5.05–5.49	6.5–6.6, 8.4–8.7	2.32–5.70	5.58–8.72
BW (GHz)	1.96	1.2	2	0.44	0.1, 0.3	3.38	3.14
Reflection Co-efficient (dB)	−29	−43	−63	−38.7	−68.317, −33.218	−45	−44.444
Resonance Freq. (GHz)	7.76	6.18	6	5.27	6.64, 8.568	5.1	8.26
Max. Gain (dB)	4.7	5	2.6	9.0828	7.015, 5.312	4	4.69
Max. Efficiency (%)	N/A	N/A	95	N/A	N/A	84	95.52

V. CONCLUSION

This study presents a triangular microstrip patch antenna optimized for C-band (5.58–8.72 GHz) communication systems which overlaps with commonly referenced portions of the C-band used for satellite uplink/downlink and fixed-wireless/backhaul services. The wide instantaneous bandwidth (≈ 3.14 GHz) enables support for multiple C-band sub-allocations and potential fixed-wireless/5G backhaul channels within this range, making the antenna suitable for hybrid satellite–terrestrial applications that require wideband operation and frequency agility. The design overcomes conventional limitations of narrow bandwidth, low efficiency, and poor impedance matching. Optimization of patch geometry and feed achieves 3.14 GHz bandwidth, 95.52% efficiency, −44.44 dB return loss, and VSWR of 1.01 at 8.26 GHz resonance. The antenna exhibits gain and directivity of 4.69 dB and 4.965 dBi, respectively, ensuring efficient energy transmission. With enhanced bandwidth, high efficiency, and stable impedance matching, the proposed

antenna offers a robust solution for satellite, 5G, and next-generation wireless communication systems.

REFERENCES

- [1] Y. Rahmat-Samii and A. C. Densmore, "Technology Trends and Challenges of Antennas for Satellite Communication Systems," in *IEEE Transactions on Antennas and Propagation*, vol. 63, no. 4, pp. 1191–1204, April 2015, doi: 10.1109/TAP.2014.2366784.
- [2] B. Mishra, R. K. Verma, N. Yashwanth, and R. K. Singh, "A review on microstrip patch antenna parameters of different geometry and bandwidth enhancement techniques," *International Journal of Microwave and Wireless Technologies*, vol. 14, no. 5, pp. 652–673, Aug. 2021, doi: 10.1017/s1759078721001148.
- [3] N. Heydarishahreza, T. Han and N. Ansari, "Spectrum Sharing and Interference Management for 6G LEO Satellite-Terrestrial Network Integration," in *IEEE Communications Surveys & Tutorials*, doi: 10.1109/COMST.2024.3507019.
- [4] Merino-Fernandez, S. L. Khemchandani, J. Del Pino, and J. Saiz-Perez, "Phased array antenna analysis workflow applied to gateways for LEO satellite communications," *Sensors*, vol. 22, no. 23, p. 9406, Dec. 2022, doi: 10.3390/s22239406.
- [5] O. Kodheli et al., "Satellite Communications in the New Space Era: A Survey and Future Challenges," in *IEEE Communications Surveys &*

Tutorials, vol. 23, no. 1, pp. 70-109, Firstquarter 2021, doi: 10.1109/COMST.2020.3028247.

- [6] K. Gupta, S. Verma and S. Khlid, "Compact Notch Loaded Monopole Antenna for C-Band Applications," *2020 International Conference on Electrical and Electronics Engineering (ICE3)*, Gorakhpur, India, 2020, pp. 530-533, doi: 10.1109/ICE348803.2020.9122784.
- [7] V. Verma, P. K. Verma, R. K. Maurya, S. Srivastava and I. Kaur, "Design & Simulation of Low Profile Wide Band Circularly Polarized Patch Antenna," *2021 IEEE Indian Conference on Antennas and Propagation (InCAP)*, Jaipur, Rajasthan, India, India, 2021, pp. 208-211, doi: 10.1109/InCAP52216.2021.9726185.
- [8] R. Sultana, A. J. Islam and M. A. Rahaman, "Design of a Truncated Circular Microstrip Patch Antenna with Slot for GPS Application," *2018 International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE)*, Gazipur, Bangladesh, 2018, pp. 1-4, doi: 10.1109/ICAEEE.2018.8643006.
- [9] M. Abdullah-Al-Mamun, S. Datto and M. R. Billah, "Inset Fed PBG Substrate with DGS Slotted Rectangular Microstrip Patch Antenna Design for C-Band Satellite Applications," *2021 International Conference on Automation, Control and Mechatronics for Industry 4.0 (ACMI)*, Rajshahi, Bangladesh, 2021, pp. 1-6, doi: 10.1109/ACMI53878.2021.9528226.
- [10] R. P, S. S, L. J. J, V. G, G. A and A. M, "Low-Profile Dual-Band Antenna for C-Band and X-Band Wireless Communication," *2025 3rd International Conference on Device Intelligence, Computing and Communication Technologies (DICCT)*, Dehradun, India, 2025, pp. 356-360, doi: 10.1109/DICCT64131.2025.10986446.
- [11] B. Nassima, B. Wahiba and A. Mehadji, "New S / C band Monopole antenna design for satellite application," *2021 IEEE International Conference on Design & Test of Integrated Micro & Nano-Systems (DTS)*, Sfax, Tunisia, 2021, pp. 1-6, doi: 10.1109/DTS52014.2021.9497993.